



Islamic University of Technology
Department of Computer Science and Engineering

Lab 6: Shortest Path

CSE 4404: Algorithms Lab
Summer 2024-25

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Task A. Logistics Network

Time Limit: 2 seconds | Memory Limit: 256 MB

A logistics company is expanding its network of n warehouses. Warehouses open one at a time. Between every ordered pair of warehouses there is a direct shipping route with a known cost. An *open* warehouse may serve as a transit point: a shipment from warehouse i to warehouse j may pass through any sequence of *currently open* intermediate warehouses.

The **shipping cost** from i to j along a route is the *sum* of costs of the individual legs. The *optimal cost* from i to j is the minimum shipping cost over all valid routes (using only currently open intermediaries).

After each warehouse opens, output the sum of optimal costs over all *ordered* pairs (i, j) of currently open warehouses with $i \neq j$. (If only one warehouse is open, this sum is 0.)

Input Format

The first line contains an integer n — the number of warehouses.

The next n lines give the cost matrix. The j -th number on the i -th line is the direct shipping cost from warehouse i to warehouse j . The cost from any warehouse to itself is 0.

The last line contains n distinct integers $x_1, x_2, \dots, x_n$ — the order in which warehouses open.

Output Format

Print n space-separated integers. The i -th integer is the total optimal cost over all ordered pairs of open warehouses immediately after warehouse x_i opens. If x_i is the first warehouse to open, print 0.

Constraints

- $1 \leq n \leq 500$
- $1 \leq \text{cost of each direct route} \leq 10^5$
- The cost from any warehouse to itself is 0.

Examples

Sample Input	Sample Output
3 0 2 9 1 0 5 4 3 0 1 2 3	0 3 22
3 0 5 2 3 0 1 8 6 0 2 3 1	0 7 25

4 0 1 10 3 2 0 4 8 7 5 0 1 6 3 2 0 3 1 4 2	0 17 24 41
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Notes

In the first sample, warehouses open in order 1, 2, 3.

After warehouse 1: only one warehouse is open, no pairs exist, total = 0.

After warehouse 2: the only routes are 1→2 with cost 2 and 2→1 with cost 1. Total = 3.

After warehouse 3: consider 1→3. The direct route costs 9. But going 1→2→3 costs $2 + 5 = 7$, which is cheaper. So the optimal cost is 7. For 3→1: direct costs 4, via warehouse 2 costs $3 + 1 = 4$ — same. All six pair costs are 2, 7, 1, 5, 4, 3, giving total = 22.

In the second sample, warehouses open in order 2, 3, 1.

After warehouse 2: total = 0.

After warehouse 3: the only routes are 2→3 with cost 1 and 3→2 with cost 6. Total = 7.

After warehouse 1: using all three warehouses, all six pair costs are 5, 2, 3, 1, 8, 6, giving total = 25.

Task B. Data Center Disaster

Time Limit: 1 second | Memory Limit: 512 MB

It’s July 4202 and the cables connecting the data centers of a country were destroyed — due to excessive rain or, strangely enough, water dropped from helicopters. Due to a sudden drop in foreign remittance, the government can only afford to repair a subset of the cables — just enough to ensure every data center can communicate with every other, at minimum cost.

Unfortunately, the engineer assigned to create the repair plan holds a BSc. in Theatre and Performance Studies and has long forgotten how to do math. So he has come to you for help in calculating the minimum budget for the project.

Input Format

The first line contains two integers n and m ($1 \leq n \leq 10^5$, $1 \leq m \leq 2 \cdot 10^5$) — the number of data centers and cables.

Each of the next m lines contains three integers a , b , and c ($1 \leq a, b \leq n$, $1 \leq c \leq 10^9$) — indicating that there is a cable between data center a and data center b , and repairing it will cost c .

All cables are bidirectional, and each connects two different data centers. There is at most one cable between any two data centers.

Output Format

Print one integer — the minimum total cost to repair enough cables so that every data center is reachable from every other, directly or indirectly.

If it is impossible to connect all data centers, print IMPOSSIBLE.

Examples

Sample Input	Sample Output
5 6 1 2 3 2 3 5 2 4 2 3 4 8 5 1 7 5 4 4	14
3 1 1 2 5	IMPOSSIBLE

Notes

In the first sample, the optimal repair set is 2→4 (cost 2), 1→2 (cost 3), 5→4 (cost 4), and 2→3 (cost 5), totalling 14. In the second sample, data center 3 has no cables, so full connectivity is impossible.

Task C. Priority Cables

Time Limit: 1 second | Memory Limit: 512 MB

Following the cable disaster of July 4202, the government has drawn up a list of *critical* cables that must be repaired no matter what — perhaps because they connect particularly important data centers, or perhaps because the minister’s nephew owns the repair company.

The remaining cables may or may not be repaired. As before, the goal is to restore full connectivity across all data centers at minimum total cost.

Help the BSc. Theatre graduate determine the minimum repair budget, keeping in mind that every critical cable must be included in the plan.

Input Format

The first line contains two integers n and m ($1 \leq n \leq 10^5$, $1 \leq m \leq 2 \cdot 10^5$) — the number of data centers and cables.

Each of the next m lines contains four integers a , b , c , and t ($1 \leq a, b \leq n$, $1 \leq c \leq 10^9$, $t \in \{0, 1\}$) — a cable between data centers a and b with repair cost c . If $t = 1$ the cable is *critical* and must be repaired; if $t = 0$ it is optional.

All cables are bidirectional, and each connects two different data centers. There is at most one cable between any two data centers.

Output Format

Print one integer — the minimum total repair cost of a spanning tree that includes all critical cables.

If the critical cables already form a cycle, or if it is impossible to connect all data centers even using every available cable, print IMPOSSIBLE.

Examples

Sample Input	Sample Output
5 7 1 2 3 1 2 3 5 0 2 4 2 1 3 4 8 0 5 1 7 0 5 4 4 0 3 5 6 0	14
3 3 1 2 4 1 2 3 3 1 1 3 2 1	IMPOSSIBLE

Notes

In the first sample, the two critical cables $1 \rightarrow 2$ (cost 3) and $2 \rightarrow 4$ (cost 2) must be included. They form no cycle, so the remaining two edges needed to connect 5 nodes are chosen optimally from the optional cables: $5 \rightarrow 4$ (cost 4) and $2 \rightarrow 3$ (cost 5), giving total $3 + 2 + 4 + 5 = 14$.

In the second sample, the three critical cables $1 \rightarrow 2$, $2 \rightarrow 3$, and $1 \rightarrow 3$ together form a cycle, so the answer is IMPOSSIBLE.

Marks Distribution

Task	Marks
Task A	40%
Task B	30%
Task C	30%

Practice Problems

Problem	Links
Road Reparation	CSES: https://cses.fi/problemset/task/1675
Building Roads	CSES: https://cses.fi/problemset/task/1666
Edges in MST	Codeforces: https://codeforces.com/problemset/problem/160/D
Minimum Spanning Tree	Codeforces: https://codeforces.com/problemset/problem/17/B